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Summary

Dr. Cui is currently an associate professor in Tongji University, Shanghai, China. He has made significant contributions to the development of stochastic modelling of tropical cyclones hazards, analytical and experimental methods about non-Gaussian turbulence effects on flexible structure, nonlinear aerodynamic effects on long-span structures. His contributions have been recognized through the 2019 American Association for Wind Engineering (AAWE) Best Paper Award, 2019 Shanghai Pujiang Talents award, 2016 Engineering Mechanics Institute Conference Student Paper Competition and Excellent Paper Awards for Young Scholars in two wind engineering conferences. Cui has authored 34 articles published in highly respected journals of structures/wind engineering/fluid-structure interaction, and has carried out 6 funded research projects related to structural wind engineering.

Education

Northeastern University, Boston, MA, USA, *Ph.D. in Civil Engineering* 2012.08 - 2017.05
Tongji University, Shanghai, China, *M.S. in Civil Engineering* 2009.09 - 2012.03
Tongji University, Shanghai, China, *B.S. in Civil Engineering* 2005.09 - 2009.07

Selected Journal Paper (* denotes corresponding author)

[J1] Cui W, Zhao L*. Using forced-motion tests to predict nonlinear flutter responses of sectional bridge model via complexification-averaging method: W. Cui, L. Zhao[J]. Nonlinear Dynamics, 2026, 114(3): 224.

[J2] Ma T, Gao T T, Cui W*, et al. Encoding Cumulation to Learn Perturbative Nonlinear Oscillatory Dynamics[J]. Advanced Science, 2026: e19707.

[J3] Zhang L, Cui W*, Zhao L. Characteristic analysis of occurrence probability for vortex-induced vibration of long-span bridge based on health monitoring system[J]. Reliability Engineering & System Safety, 2025: 111516.

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Long Term Trend and Ambient Interference[J]. Structural Control and Health Monitoring, 2023, 2023(1): 6485040.

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Award

- **2019 Best Journal Paper Award**, *AAWE*, A new stochastic formulation for synthetic hurricane simulation over the North Atlantic Ocean[J]. Engineering Structures, 2019, 199: 109597.
- **Pujiang Talents**, *Shanghai Scientific Committee*, 2019
- **Winners in the Student Paper Competition**, *EMI-PMC 2016*, Statistical modeling of Hurricane over North Atlantic Ocean